

can't AI data centers re-use the same water?

Posted by u/BottleSad505 • 0 points • 9 comments

I've heard ppl say that every AI prompt uses like a gallon of water (or less idk) and that water is just lost. But have we forgotten how the water cycle works? It rains, water goes back to the ocean/lakes, and once hot enough, the surface of the water evaporates and goes back into the skies?

other things are

But aside from that, I am studying marine engineering (and only a student, so my logic might be wrong, so correct me by all means), and on board of fishing vessels, I have learnt that engineers use seawater as coolants to lower the temperature of the engine, and they use antifreeze for the water.

So basically, what I am asking is-- Do AI centers use as much water as people claim? And the jokes about "AI is emptying all the water from lakes!!" Is it really that true? I mean, there are other things that are also bad, if not worse, right?

I just started thinking about it too deeply, so if I am completely off the rails, do let me know before I spread out false information lol

Comments

u/DiogenesKuon • 1 points • 2026-03-09 00:02 UTC

The water usage by AI is vastly overstated most of the time. While data centers do use a lot of water it's a tiny amount of the total water we use for other activities. Just in the US we use more water keeping golf courses green than all data centers (AI and otherwise) combined use for cooling.

u/beast4rent • 3 points • 2026-03-08 23:25 UTC

It's a slight exaggeration to say that lakes are being emptied, but messing with the water cycle and displacing water does affect bodies of water. The amount of water in a lake is based around a lake's ****natural replenishment rate****. Taking the water out of the lake and using it messes with that equilibrium.

Consider this; if you take half of the water in a lake, walk 10 miles to another lake and dump all the water in that one. The water will still evaporate and rain to the same basin. Will all the water rain back into lake #1? How will the wildlife in lake #1 react? Would having less water affect the ecosystem? All this to say, displacement of water isn't neutral.

u/Alesus2-0 • 5 points • 2026-03-08 23:22 UTC

The water consumed by data centres is largely used for evaporative cooling. The water is evaporated to remove heat from the facility and then vented into the outside air. Obviously, that vapour doesn't cease to exist, but it also doesn't automatically condense itself in a location as convenient as the original water source.

In principle, it would certainly be possible to capture this vapour, cool it until it condenses and then reuse it. But that would involve significant extra cost and complexity. The big appeals of evaporative cooling are that it is cheap and simple.

u/Lumpy-Notice8945 • 2 points • 2026-03-08 23:19 UTC

> I've heard ppl say that every AI prompt uses like a gallon of water (or less idk)

That's just false, it's way less, like some drops or a teaspoon at most. Look up some actual reliable sources that try to estimate it, it's a big range but none will claim a gallon or even close to that.

A whole server farm, aka datacenter has an AC build in that cools the whole building, some of these evaporate tap/potable water, that's it.

So you can't reuse that evaporated water because well it's evaporated and spread into the atmosphere as a gas. It's not "lost forever" it's not destroyed in a sense that the molecules don't exist anymore or that it's split into hydrogen and oxygen by some reaction it's just out of the tap and into the air or sewer.

u/ForScale • 6 points • 2026-03-08 23:17 UTC

They evaporate it into the environment. To capture and reuse it would cost them more money.

u/BottleSad505 • 1 point • 2026-03-08 23:20 UTC

but where are data centers normally located? Because I thought it would normally be near a water source, right? Or am I tripping for thinking that

u/Indemnity4 • 2 points • 2026-03-09 05:28 UTC

They need access to a lot of electricity. It's their main cost. For this reason they are usually constructed close to power generators or at least high transmission power lines.

Then they want access to communication lines. There are giant trunk lines of fibre optic swinging across the country.

Water is important but it's probably 3rd or maybe 5th down that list. You need access to construction companies and maybe even skilled workers, probably a decent fire department, some cheap land.

u/BottleSad505 • 1 point • 2026-03-09 08:31 UTC

Oh I see, thank you for explaining!

u/aRabidGerbil • 4 points • 2026-03-08 23:28 UTC

They're located wherever land is cheap and they can get the local government to give them tax breaks